

only) and since all users of the SAR (Spatial Augmented Reality) setup can see each other, they can easily collaborate.

Though we have mentioned quite a number of displays that are used for AR, there are others which are not as popular. Let us now go to devices and methods used for inputting data into the AR systems.

Input hardware for Interaction

“We think basically you watch television to turn your brain off, and you work on your computer when you want to turn your brain on.” - Steve Jobs

While opinions may vary greatly when it comes to Steve Jobs, it is pretty difficult to defy the logic contained in the sentence quoted above. Interaction is the main difference between two things that look just alike: a television and a PC. Both of them have got a display system but the former does not allow you to play as much with it as the latter does. While a TV would allow you to change channels and turn up or down the volume, a PC allows you to do much more (even while speaking of simply entertainment).

AR is one of the most active areas of research in computing but strangely AR doesn't require much interaction. In fact much of the interaction occurring within different types of AR implementations is automated to a level that you do not have to command the system once it has started working. A lot of research is directed towards minimizing whatever interaction you do with the system. For instance, if you were to wear a HMD, you would not want to keep pressing buttons to command the HMD to display what is on the right, instead you would turn right and the HMD would do the job of augmentation of the environment to your right. Such an experience does not require you to constantly command the system. The system detects your movements and based on prediction of what you would want, changes the view.

But not all AR implementations are used or are intended to be used without any interaction. Take for example the tourism AR applications you could install on your phone. Some of them do require you to tell them what information you want augmented into the real view. In the famous AR browser application 'Layar', you would have to first select the 'layer' of information which should provide the augmentation. Other methods of manual input involve verbal commands that can be given to an AR system.

Let us again take the example of HMD. In most cases a HMD would cover the entire viewing area in front of your eye. In such a case it becomes very cumbersome to command the system using controls such as buttons.